

ASSIGNMENT-1

04/07/26-MINI PROJECT ON NORMALIZATION

1. Normalize the following table into 1 Normal Form

Original Table

Member_Id	First_Name	Last_Name	Hobbies
101	Jayson	Mark	Cricket,Swimming,Football
102	Ram	Ganesh	Swimming,Running,Music
103	Raj	Kishore	Dancing,Singing,Running

1NF

Member_Id	First_Name	Last_Name	Hobby
101	Jayson	Mark	Cricket
101	Jayson	Mark	Swimming
101	Jayson	Mark	Football
102	Ram	Ganesh	Swimming
102	Ram	Ganesh	Running
102	Ram	Ganesh	Music
103	Raj	Kishore	Dancing
103	Raj	Kishore	Singing
103	Raj	Kishore	Running

2. Normalize the following table into 2nd Normal Form

Original Table

Empno	Training	Training_Date	Dept
101	Oracle SQL	12-Aug-2015	TT
101	Java	21-Aug-2015	BU
102	Oracle SQL	18-Sep-2014	TT

2NF

Employee Table

Empno Dept

101	TT
102	TT

Training Table

Empno	Training	Training_Date
101	Oracle SQL	12-Aug-2015
101	Java	21-Aug-2015
102	Oracle SQL	18-Sep-2014

3. Normalize the following table into 3rd Normal Form

Original Table

Member_Id	First_Name	Last_Name	Sports	Fees
101	Rajesh	Chand	Cricket	100
102	Jayesh	Raj	Hockey	80
103	Mark	Dorson	Football	90

3NF

Member Table

Member_Id	First_Name	Last_Name	Sports
101	Rajesh	Chand	Cricket
102	Jayesh	Raj	Hockey
103	Mark	Dorson	Football

Sports Fee Table

Sports Fees

Cricket 100
Hockey 80
Football 90

ASSIGNMENT-2

> [07/07/2026: RETREIVING THE DATA USING SQL SELECT STATEMENT]

1. Determine the Structure of Dept Table and its Contents.

```
Command Prompt - sqlplus / X + v

SQL> create table dept(
  2 deptno int primary key,
  3 dname varchar2(14),
  4 location varchar2(13)
  5 );

Table created.

SQL> insert into dept values(10,'ACCOUNTING','NEW YORK');

1 row created.

SQL> insert into dept values(20,'RESEARCH','DALLAS');

1 row created.

SQL> insert into dept values(30,'SALES','CHICAGO');

1 row created.

SQL> insert into dept values(40,'OPERATIONS','BOSTON');

1 row created.

SQL> commit;

Commit complete.

SQL> desc dept;
Name                               Null?    Type
-----
DEPTNO                             NOT NULL NUMBER(38)
DNAME                              VARCHAR2(14)
LOCATION                             VARCHAR2(13)

SQL> select * from dept;

DEPTNO DNAME          LOCATION
-----
10 ACCOUNTING    NEW YORK
20 RESEARCH     DALLAS
30 SALES        CHICAGO
40 OPERATIONS   BOSTON
```

2. Determine the Structure of Emp Table and its Contents.

```

SQL> create table emp(
  2 empno int primary key,
  3 ename varchar2(10),
  4 job varchar2(10),
  5 salary int,
  6 deptno int
  7 );

Table created.

SQL> insert into emp values(7369,'SMITH','CLERK',800,20);

1 row created.

SQL> insert into emp values(7499,'ALLEN','SALESMAN',1600,30);

1 row created.

SQL> insert into emp values(7566,'ARUN','MANAGER',2975,20);

1 row created.

SQL> insert into emp values(7788,'KIRAN','ANALYST',3000,20);

1 row created.

SQL> insert into emp values(7839,'MEGHA','PRESIDENT',5000,10);

1 row created.

SQL> desc emp;
Name                               Null?    Type
-----
EMPNO                               NOT NULL NUMBER(38)
ENAME                               VARCHAR2(10)
JOB                                 VARCHAR2(10)
SALARY                             NUMBER(38)
DEPTNO                             NUMBER(38)

SQL> select * from emp;

  EMPNO  ENAME      JOB              SALARY   DEPTNO
-----
  7369  SMITH      CLERK              800        20
  7499  ALLEN     SALESMAN          1600        30
  7566  ARUN      MANAGER           2975        20
  7788  KIRAN     ANALYST           3000        20
  7839  MEGHA     PRESIDENT         5000        10

```

3. Display the Ename and Deptno from Emp table whose Empno is 7788.

```

SQL> select ename,deptno
  2 from emp
  3 where empno=7788;

ENAME      DEPTNO
-----
KIRAN      20

SQL> |

```

ASSIGNMENT-3

>[08/07/2026:RESTRICTING AND SORTING DATA]

```

SQL> create table emp1
2 (
3 empno int,
4 ename varchar(20),
5 job varchar(20),
6 mgr int,
7 hiredate date,
8 sal int,
9 comm int,
10 deptno int
11 );

Table created.

SQL> insert into emp1 values(7369,'smitha','clerk',7902,'17-dec-80',800,null,20);
1 row created.

SQL> insert into emp1 values(7499,'allen','salesman',7698,'20-feb-81',1600,300,30);
1 row created.

SQL> insert into emp1 values(7566,'jones','manager',7839,'02-apr-81',2975,null,20);
1 row created.

SQL> insert into emp1 values(7698,'blake','manager',7839,'01-may-81',2850,null,30);
1 row created.

SQL> insert into emp1 values(7788,'scott','analyst',7566,'19-apr-87',3000,null,20);
1 row created.

SQL> insert into emp1 values(7839,'king','president',null,'17-nov-81',5000,null,10);
1 row created.

SQL> insert into emp1 values(7844,'tarun','salesman',7698,'08-sep-81',1500,0,30);
1 row created.

SQL> insert into emp1 values(7900,'james','clerk',7698,'03-dec-81',950,null,30);
1 row created.

SQL> insert into emp1 values(7934,'miller','clerk',7782,'23-jan-82',1300,null,10);
1 row created.

```

1. Display the Ename, Sal, Comm from Emp table who earn commission and sort the records in descending order of Salary and Comm. Use column's numeric position in the ORDER BY clause.

```

SQL> select ename, sal, comm
2 from emp1
3 where comm is not null
4 order by 2 desc, 3 desc;

```

ENAME	SAL	COMM
allen	1600	300
tarun	1500	0

2. The HR department needs a query to display all unique job codes from the EMP table.

```

SQL> select distinct job
2 from emp1;

```

```

JOB
-----
clerk
salesman
manager
analyst
president

```

3. The HR department wants more descriptive column headings for its report on employees. Name the column headings Emp #, Employee, Job, and Hire Date, respectively by giving Column Aliases.

```

SQL> select empno "emp #",
2 ename "employee",
3 job "job",
4 hiredate "hire date"
5 from emp1;

```

emp #	employee	job	hire date
7369	smitha	clerk	17-DEC-80
7499	allen	salesman	20-FEB-81
7566	jones	manager	02-APR-81
7698	blake	manager	01-MAY-81
7788	scott	analyst	19-APR-87
7839	king	president	17-NOV-81
7844	tarun	salesman	08-SEP-81
7900	james	clerk	03-DEC-81
7934	miller	clerk	23-JAN-82

9 rows selected.

4. The HR department has requested a report of all employees and their job IDs. Display the last name concatenated with the Job ID (separated by a comma and a space) and name the column Employee and Title by giving Column Aliases.

```
SQL> select ename || ', ' || job "employee and title"
2 from emp1;

employee and title
-----
smitha, clerk
allen, salesman
jones, manager
blake, manager
scott, analyst
king, president
tarun, salesman
james, clerk
miller, clerk

9 rows selected.
```

5. Create a query to display all the data from EMP table. Separate each column output by a comma. Ename, Job, Hiredate, Mgr. Name the column title THE_OUTPUT.

```
SQL> select ename || ', ' || job || ', ' || hiredate || ', ' || mgr "the_output"
2 from emp1;

the_output
-----
smitha,clerk,17-DEC-80,7902
allen,salesman,20-FEB-81,7698
jones,manager,02-APR-81,7839
blake,manager,01-MAY-81,7839
scott,analyst,19-APR-87,7566
king,president,17-NOV-81,
tarun,salesman,08-SEP-81,7698
james,clerk,03-DEC-81,7698
miller,clerk,23-JAN-82,7782

9 rows selected.
```

6. Create a report to display the ename, job, and Hiredate for the employees whose name is SCOTT or TURNER. Order the query in ascending order by hiredate.

```
SQL>
SQL> select ename, job, hiredate
2 from emp1
3 where ename in ('scott','turner')
4 order by hiredate;

ENAME                JOB                HIREDATE
-----
turner               salesman          08-SEP-81
scott                analyst          19-APR-87
```

7. Display the ename and department number of all employees in departments 20 or 30 in ascending alphabetical order by ename.

```
SQL> select ename, deptno
2 from emp1
3 where deptno in (20,30)
4 order by ename;

ENAME                DEPTNO
-----
allen                30
blake                30
james                30
jones                20
scott                20
smitha              20
tarun                30

7 rows selected.
```

8. Modify the previous query to display the last name and salary of employees who earn between 2000 and 3000 and are in department 20 or 30. Label the columns Employee and Monthly Salary.

```
SQL> select ename "employee",
2 sal "monthly salary"
3 from emp1
4 where sal between 2000 and 3000
5 and deptno in (20,30);

employee            monthly salary
-----
jones                2975
blake                2850
scott                3000
```

9. The HR department needs a report that displays the last name and hire date for all employees who were hired in 1981.

```
SQL> select ename, hiredate
  2   from emp1
  3  where hiredate between '01-jan-1981' and '31-dec-1981';
```

ENAME	HIREDATE
allen	20-FEB-81
jones	02-APR-81
blake	01-MAY-81
king	17-NOV-81
tarun	08-SEP-81
james	03-DEC-81

6 rows selected.

10. Display the Ename, Sal of employees who earn more than an amount the user specifies after a Prompt.

```
SQL> select ename, sal
  2   from emp1
  3  where sal > &salary;
Enter value for salary: 2000
old  3: where sal > &salary
new  3: where sal > 2000
```

ENAME	SAL
jones	2975
blake	2850
scott	3000
king	5000

11. Create a report to display the last name and job title of all employees who do not have a manager.

```
SQL> select ename, job
  2   from emp1
  3  where mgr is null;
```

ENAME	JOB
king	president

12. Create a query that prompts the user for Manager ID and generate EMPNO, ENAME, SAL, DEPTNO. The user should have the ability to sort the records on a selected Column.

```
SQL> select empno, ename, sal, deptno
  2   from emp1
  3  where mgr = &mgr_id
  4  order by &column_name;
Enter value for mgr_id: 7698
old  3: where mgr = &mgr_id
new  3: where mgr = 7698
Enter value for column_name: ename
old  4: order by &column_name
new  4: order by ename
```

EMPNO	ENAME	SAL	DEPTNO
7499	allen	1600	30
7900	james	950	30
7844	tarun	1500	30

13. Create a query that prompts the user for a MGR and generates the empno, ename, salary, and department for that manager's employees. The HR department wants the ability to sort the report on a selected column.

```
SQL> select empno, ename, sal, deptno
  2   from emp1
  3  where mgr = &mgr
  4  order by &column_name;
Enter value for mgr: 7698
old  3: where mgr = &mgr
new  3: where mgr = 7698
Enter value for column_name: ename
old  4: order by &column_name
new  4: order by ename
```

EMPNO	ENAME	SAL	DEPTNO
7499	allen	1600	30
7900	james	950	30
7844	tarun	1500	30

14. Display all employee last names in which the third letter of the name is A.

```
SQL> select ename
  2   from emp1
  3  where ename like '__a%';

ENAME
-----
blake
```

15. Display the last name of all employees who have both an A and an S in their ename.

```
SQL> select ename
  2   from emp1
  3  where ename like '%a%'
  4    and ename like '%s%';

ENAME
-----
smitha
james
```

16. Display the Ename, Job, Sal for all employees whose jobs are CLERK and whose salary is 800 or 950 or 1300.

```
SQL> select ename, job, sal
  2   from emp1
  3  where job = 'clerk'
  4    and sal in (800,950,1300);

ENAME          JOB          SAL
-----
smitha         clerk          800
james          clerk          950
miller         clerk         1300
```

ASSIGNMENT-4

> [09/07/2026:USING SINGLE ROW FUNCTIONS TO CUSTOMIZE THE OUTPUT]

1. Write a query to display the current date. Label the column **Date**.

```
SQL> select sysdate "date"
  2   from dual;

date
-----
11-JUL-26
```

2. The HR department needs a report to display the employee number, last name, salary, and salary increased by **15.5%** (expressed as a whole number) for each employee. Label the column **New Salary**.

```
SQL> select empno,
  2          ename,
  3          sal,
  4          round(sal*1.155) "new salary"
  5   from emp1;

EMPNO  ENAME          SAL  new salary
-----
7369  smith            800      924
7499  allen           1600     1848
7566  jones            2975     3436
7698  blake            2850     3292
7788  scott            3000     3465
7839  king             5000     5775
7844  turner           1500     1733
7900  james            950     1097
7934  miller          1300     1502

9 rows selected.
```

3. Modify the previous query to add a column alias that subtracts the old salary from the new salary. Label the column **Increase**.

```
SQL> select empno,
2      ename,
3      sal,
4      round(sal*1.155) "new salary",
5      round(sal*1.155)-sal "increase"
6  from emp1;
```

EMPNO	ENAME	SAL	new salary	increase
7369	smith	800	924	124
7499	allen	1600	1848	248
7566	jones	2975	3436	461
7698	blake	2850	3292	442
7788	scott	3000	3465	465
7839	king	5000	5775	775
7844	turner	1500	1733	233
7900	james	950	1097	147
7934	millar	1300	1502	202

9 rows selected.

4. Write a query that displays the **ename** (with the first letter uppercase and all other letters lowercase) and the length of the **ename** for all employees whose ename starts with the letters **J, A, or M**. Give each column an appropriate label. Sort the results by the employees' enames.

```
SQL> select initcap(ename) "employee",
2      length(ename) "length"
3  from emp1
4  where ename like 'j%'
5  or ename like 'a%'
6  or ename like 'm%'
7  order by ename;
```

employee	length
Allen	5
James	5
Jones	5
Miller	6

5. Rewrite the query so that the user is prompted to enter a letter that starts the last name. For example, if the user enters **H** when prompted for a letter, then the output should show all employees whose last name starts with the letter **H**.

```
SQL> select ename
2  from emp1
3  where ename like '&letter%';
Enter value for letter: a
old 3: where ename like '&letter%'
new 3: where ename like 'a%'

ENAME
-----
allen
```

6. The HR department wants to find the length of employment for each employee. For each employee, display the **ename** and calculate the number of months between today and the date on which the employee was hired. Label the column **MONTHS_WORKED**. Order your results by the number of months employed. Round the number of months up to the closest whole number.

```
SQL> select ename,
2      round(months_between(sysdate,hiredate)) "months_worked"
3  from emp1
4  order by round(months_between(sysdate,hiredate));
```

ENAME	months_worked
scott	471
millar	534
james	535
king	536
turner	538
blake	542
jones	543
allen	545
smith	547

9 rows selected.

7. Create a report that produces the following for each employee: **earns monthly but wants <3 times salary>**. Label the column **Dream Salaries**.


```
SQL> select ename||' earns '||sal||' monthly but wants '||sal*3 "dream salaries"
2 from emp1;

dream salaries
-----
smith earns 800 monthly but wants 2400
allen earns 1600 monthly but wants 4800
jones earns 2975 monthly but wants 8925
blake earns 2850 monthly but wants 8550
scott earns 3000 monthly but wants 9000
king earns 5000 monthly but wants 15000
turner earns 1500 monthly but wants 4500
james earns 950 monthly but wants 2850
miller earns 1300 monthly but wants 3900

9 rows selected.
```

8. . Create a query to display the last name and salary for all employees. Format the salary to be **15 characters long, left-padded with the \$ symbol**. Label the column **SALARY**.

```
SQL> select ename,
2          lpad(sal,15,'$') "salary"
3 from emp1;

ENAME
-----
salary
-----
smith
$$$$$$$$$$$$$800
allen
$$$$$$$$$$$$$1600
jones
$$$$$$$$$$$$$2975

ENAME
-----
salary
-----
blake
$$$$$$$$$$$$$2850
scott
$$$$$$$$$$$$$3000
king
$$$$$$$$$$$$$5000

ENAME
-----
salary
-----
turner
$$$$$$$$$$$$$1500
james
$$$$$$$$$$$$$950
miller
$$$$$$$$$$$$$1300

9 rows selected.
```

9. Display each employee's last name, hire date, and salary review date, which is the first Monday after six months of service. Label the column **REVIEW**. Format the dates to appear in the format similar to **"Monday, the Thirty-First of July, 2000."**

```
SQL> select ename,
2          hiredate,
3          to_char(next_day(add_months(hiredate,6),'monday'),
4                  'fmday," the "dvsp of "month, yyyy') "review"
5 from emp1;

ENAME          HIREDATE
-----
review
-----
smith          17-DEC-80
monday, the twenty-two of june, 1981
allen          20-FEB-81
monday, the twenty-four of august, 1981
jones          02-APR-81
monday, the five of october, 1981

ENAME          HIREDATE
-----
review
-----
blake          01-MAY-81
monday, the two of november, 1981
scott          19-APR-87
monday, the twenty-six of october, 1987
king          17-NOV-81
monday, the twenty-four of may, 1982

ENAME          HIREDATE
-----
review
-----
turner          08-SEP-81
monday, the fifteen of march, 1982
james          03-DEC-81
monday, the seven of june, 1982
miller          23-JAN-82
monday, the twenty-six of july, 1982
```

10. Display the last name, hire date, and day of the week on which the employee started. Label the column **DAY**. Order the results by the day of the week, starting with Monday.

```
SQL> select ename,
2         hiredate,
3         to_char(hiredate,'day') "day"
4   from emp
5   order by to_char(hiredate,'d');

ENAME          HIREDATE    day
-----
scott          19-APR-87  sunday
king          17-NOV-81  tuesday
turner        08-SEP-81  tuesday
smith         17-DEC-80  wednesday
jones         02-APR-81  thursday
james         03-DEC-81  thursday
allen         20-FEB-81  friday
blake         01-MAY-81  friday
miller        23-JAN-82  saturday

9 rows selected.
```

11. Create a query that displays the employees' last names and commission amounts. If an employee does not earn commission, show **"No Commission."** Label the column **COMM**.

```
SQL> select ename,
2         nvl(to_char(comm), 'No Commission') "comm"
3   from emp;

ENAME          comm
-----
smith          No Commission
allen          300
jones          No Commission
blake          No Commission
scott          No Commission
king           No Commission
turner         0
james          No Commission
miller         No Commission

9 rows selected.
```

12. . Create a query that displays the first eight characters of the employees' last names and indicates the amounts of their salaries with asterisks. Each asterisk signifies a thousand dollars. Sort the data in descending order of salary. Label the column **EMPLOYEES AND THEIR SALARIES**.

```
SQL> select rpad(substr(ename,1,8),8)||' '||
2         rpad('*',sal/1000,'*') "employees and their salaries"
3   from emp
4   order by sal desc;

employees and their salaries
-----
king          *****
scott         ***
jones         **
blake         **
allen         *
turner        *
miller        *
james
smith

9 rows selected.
```

13. Using the **DECODE** function, write a query that displays the grade of all employees based on the value of the column **JOB**, using the following data:

- PRESIDENT-A
- MANAGER-B
- ANALYST-C
- SALESMAN-C
- CLERK-D

```
SQL> select ename,
2         job,
3         decode(job, 'A',
4         'president', 'B',
5         'manager', 'C',
6         'analyst', 'C',
7         'salesman', 'C',
8         'clerk', 'D') "grade"
9   from emp1;
```

ENAME	JOB	g
smith	clerk	D
allen	salesman	C
jones	manager	B
blake	manager	B
scott	analyst	C
king	president	A
turner	salesman	C
james	clerk	D
miller	clerk	D

9 rows selected.

14. Rewrite the statement in the preceding exercise using the **CASE** syntax.

```
SQL> select ename,
2         job,
3         case
4         when job='president' then 'A'
5         when job='manager' then 'B'
6         when job='analyst' then 'C'
7         when job='salesman' then 'C'
8         when job='clerk' then 'D'
9         end "grade"
10  from emp1;
```

ENAME	JOB	g
smith	clerk	D
allen	salesman	C
jones	manager	B
blake	manager	B
scott	analyst	C
king	president	A
turner	salesman	C
james	clerk	D
miller	clerk	D

9 rows selected.

ASSIGNMENT-5

10/07/2026: REPORTING AGGREGATED DATA USING THE GROUP FUNCTIONS

No.	Cat	Hands-on Assignment	Topics Covered	Status
1	M	Find the highest, lowest, sum, and average salary of all employees. Label the columns	Group Functions	☐
2	M	Maximum, Minimum, Sum, and Average, respectively. Round your results to the nearest whole number.	Group Functions	☐
3	M	Modify the above query to display the minimum, maximum, sum, and average salary for each job type.	Group by Clause	☐
4	M	Write a query to display the number of people with the same job	Group Functions	☐
5	M	Determine the number of managers without listing them. Label the column Number of Managers	Group by Clause	☐
6	M	Find the difference between the highest and lowest salaries. Label the column DIFFERENCE.	Group Functions	☐
7	M	Create a report to display the manager number and the salary of the lowest-paid employee for that manager. Exclude anyone whose manager is not known. Exclude any groups where the minimum salary is \$2000 or less. Sort the output in descending order of salary.	Where Clause, Group by Clause, Having Clause, Order by Clause	☐
8	M	Create a query to display the total number of employees and, of that total, the number of employees hired in 1980, 1981, and 1982. Create appropriate column headings.	Single Row and Group Functions	☐

```
SQL> select max(sal) as "maximum",
2         min(sal) as "minimum",
3         sum(sal) as "sum",
4         round(avg(sal)) as "average"
5 from empl;

maximum      minimum      sum      average
-----
5000          800      19975      2219

SQL> select job,
2         min(sal) as "minimum",
3         max(sal) as "maximum",
4         sum(sal) as "sum",
5         round(avg(sal)) as "average"
6 from empl
7 group by job;

JOB              minimum      maximum      sum      average
-----
clerk              800        1300      3050      1017
salesman          1500        1600      3100      1550
manager           2850        2975      5825      2913
analyst           3000        3000      3000      3000
president         5000        5000      5000      5000

SQL> select job,
2         count(*) as "number of people"
3 from empl
4 group by job;

JOB              number of people
-----
clerk              3
salesman           2
manager            2
analyst            1
president          1

SQL> select count(distinct mgr) as "number of managers"
2 from empl;

number of managers
-----
5
```

```

SQL> select max(sal)-min(sal) as "difference"
2   from emp1;

difference
-----
      4200

SQL> select mgr,
2   min(sal) as "lowest salary"
3   from emp1
4   where mgr is not null
5   group by mgr
6   having min(sal) > 2000
7   order by min(sal) desc;

MGR lowest salary
-----
7566          3000
7839          2850

SQL> select count(*) as "total employees",
2   sum(case when extract(year from hiredate)=1980 then 1 else 0 end) as "1980",
3   sum(case when extract(year from hiredate)=1981 then 1 else 0 end) as "1981",
4   sum(case when extract(year from hiredate)=1982 then 1 else 0 end) as "1982"
5   from emp1;

total employees      1980      1981      1982
-----
          9          1          6          1

SQL> |

```

ASSIGNMENT-6

11/07/26:USING THE SET OPERATORS

No.	Cat	Hands-on Assignment
1	M	Create a matrix query to display the job, the salary for that job based on department number, and the total salary for that job, for departments 10, 20, and 30, giving each column an appropriate heading.
2	M	Using set operator display the DEPTNO,SUM(SAL) for each dept, JOB,SUM(SAL) for each Job and Total Salary.
3	M	Using Set Operator display the JOB and Deptno in employees working in deptno 20,10,30 in that order.

```

SQL> select job,
2   sum(case when deptno=10 then sal else 0 end) as dept10,
3   sum(case when deptno=20 then sal else 0 end) as dept20,
4   sum(case when deptno=30 then sal else 0 end) as dept30,
5   sum(sal) as total
6   from emp1
7   group by job;

```

JOB	DEPT10	DEPT20	DEPT30	TOTAL
clerk	1300	800	950	3050
salesman	0	0	3100	3100
manager	0	2975	2850	5825
analyst	0	3000	0	3000
president	5000	0	0	5000

```

SQL> select to_char(deptno) as category,
2         sum(sal) as total_salary
3   from emp1
4  group by deptno
5  union
6  select job as category,
7         sum(sal) as total_salary
8   from emp1
9  group by job
10 union
11 select 'total' as category,
12        sum(sal) as total_salary
13   from emp1;

```

CATEGORY	TOTAL_SALARY
20	6775
30	6900
10	6300
clerk	3050
salesman	3100
manager	5825
analyst	3000
president	5000
total	19975

9 rows selected.

```

SQL> select job,deptno
2   from emp1
3  where deptno=20
4  union
5  select job,deptno
6   from emp1
7  where deptno=10
8  union
9  select job,deptno
10  from emp1
11  where deptno=30;

```

JOB	DEPTNO
clerk	20
manager	20
analyst	20
president	10
clerk	10
salesman	30
manager	30
clerk	30

8 rows selected.

ASSIGNMENT-7

13/7/26:DISPLAYING DATA FROM MULTIPLE TABLES USING JOINS

1	M	Write a query for the HR department to produce the addresses of all the departments. Use the EMP and DEPT tables. Show the EMPNO, ENAME,SAL, DNAME and LOC in the output. Use a NATURAL JOIN to produce the results.	Natural Join
2	M	The HR department needs a report of all employees. Write a query to display the JOB,MGR,SAL,COMM,DNAME of employees whose JOB is SALESMAN.	Equi Join
3	M	The HR department needs a report of employees in LOC DALLAS. Display the ENAME, job, DEPTNO, and DNAME for all employees who work in DALLAS.	Equi Join
4	M	Create a report to display employees' ename and employee number along with their manager's name and manager number. Label the columns Employee, Emp#, Manager, and Mgr#, respectively.	Self Join
5	M	Modify the previous Query to display all employees including King, who has no manager. Order the results by the employee number.	Outer Join
6	M	The HR department needs a report on job grades and salaries. To familiarize yourself with the SALGRADE table, first show the structure of the SALGRADE table. Then create a query that displays the name, job, department name, salary, and grade for all employees.	Non-Equi Join
7	M	Display the ENAME,DNAME of all the employees. Also display those department name which do not have any employees working.	Outer Join
8	M	The HR department needs to find the names and hire dates for all employees who were hired before their managers, along with their managers' names and hire dates.	Self Join with Outer Join
9	M	Display the EMPNO,ENAME,DNAME,LOC of those employees who are working as CLERK. Use the USING clause.	USING Clause
10	M	Display the ENAME,SAL,MGR,DNAME of employees whose salary is more than 2000. Use the ON clause.	On Clause
11	M	Display the EMPNO,ENAME,JOB,DEPTNO,DNAME,LOC of employees. Use LEFT OUTER JOIN.	LEFT OUTER JOIN
12	M	Display the ENAME,DNAME of employees. Use RIGHT OUTER JOIN.	RIGHT OUTER JOIN
13	M	Display the EMPNO,DNAME,LOC of employees. Use FULL OUTER JOIN.	FULL OUTER JOIN

```
SQL> select empno,ename,sal,dname,location
2 from empl
3 natural join dept;
```

EMPNO	ENAME	SAL	DNAME	LOCATION
7369	smith	800	RESEARCH	DALLAS
7499	allen	1600	SALES	CHICAGO
7566	jones	2975	RESEARCH	DALLAS
7698	blake	2850	SALES	CHICAGO
7788	scott	3000	RESEARCH	DALLAS
7839	king	5000	ACCOUNTING	NEW YORK
7844	turner	1500	SALES	CHICAGO
7900	james	950	SALES	CHICAGO
7934	millar	1300	ACCOUNTING	NEW YORK

9 rows selected.

```
SQL> select job,mgr,sal,comm,dname
2 from empl e,dept d
3 where e.deptno=d.deptno
4 and job='salesman';
```

JOB	MGR	SAL	COMM	DNAME
salesman	7698	1600	300	SALES
salesman	7698	1500	0	SALES

```
SQL> select ename,job,e.deptno,dname
2 from empl e,dept d
3 where e.deptno=d.deptno
4 and location='DALLAS';
```

ENAME	JOB	DEPTNO	DNAME
smith	clerk	20	RESEARCH
jones	manager	20	RESEARCH
scott	analyst	20	RESEARCH

```
SQL> select e.ename as employee,
2 e.empno as "emp#",
3 m.ename as manager,
4 m.empno as "mgr#"
5 from empl e,empl m
6 where e.mgr=m.empno;
```

EMPLOYEE	emp#	MANAGER	mgr#
scott	7788	jones	7566
allen	7499	blake	7698
turner	7844	blake	7698
james	7900	blake	7698
jones	7566	king	7839
blake	7698	king	7839

6 rows selected.

```
SQL> select e.ename as employee,
2 e.empno as "emp#",
3 m.ename as manager,
4 m.empno as "mgr#"
5 from empl e
6 left join empl m
7 on e.mgr=m.empno
8 order by e.empno;
```

EMPLOYEE	emp#	MANAGER	mgr#
smith	7369		
allen	7499	blake	7698
jones	7566	king	7839
blake	7698	king	7839
scott	7788	jones	7566
king	7839		
turner	7844	blake	7698
james	7900	blake	7698
millar	7934		

9 rows selected.

```
SQL> select ename,
2 job,
3 dname,
4 sal,
5 grade
6 from empl e,salgrade s,dept d
7 where e.deptno=d.deptno
8 and sal between losal and hisal;
```

ENAME	JOB	DNAME	SAL	GRADE
millar	clerk	ACCOUNTING		1300
king	president	ACCOUNTING		5000
smith	clerk	RESEARCH	800	1
jones	manager	RESEARCH	2975	4
scott	analyst	RESEARCH	3000	4
james	clerk	SALES	950	1
allen	salesman	SALES	1600	3
turner	salesman	SALES	1500	3
blake	manager	SALES	2850	4

9 rows selected.

```
SQL> select ename,dname
2 from dept d
3 left join empl e
4 on d.deptno=e.deptno;
```

ENAME	DNAME
smith	RESEARCH
allen	SALES
jones	RESEARCH


```

blake      SALES
scott      RESEARCH
king       ACCOUNTING
turner     SALES
james      SALES
miller     ACCOUNTING
           OPERATIONS

```

10 rows selected.

```

SQL> select e.ename,
2         e.hiredate,
3         m.ename,
4         m.hiredate
5   from emp1 e
6  left join emp1 m
7    on e.mgr=m.empno
8  where e.hiredate<m.hiredate;

```

ENAME	HIREDATE	ENAME	HIREDATE
allen	20-FEB-81	blake	01-MAY-81
jones	02-APR-81	king	17-NOV-81
blake	01-MAY-81	king	17-NOV-81

```

SQL> select empno,ename,dname,location
2   from emp1
3  join dept
4  using(deptno)
5  where job='clerk';

```

EMPNO	ENAME	DNAME	LOCATION
7934	miller	ACCOUNTING	NEW YORK
7369	smith	RESEARCH	DALLAS
7900	james	SALES	CHICAGO

```

SQL> select ename,sal,mgr,dname
2   from emp1 e
3  join dept d
4  on e.deptno=d.deptno
5  where sal>2000;

```

ENAME	SAL	MGR	DNAME
king	5000		ACCOUNTING
jones	2975	7839	RESEARCH
scott	3000	7566	RESEARCH
blake	2850	7839	SALES

```

SQL> select empno,ename,job,e.deptno,dname,location
2   from emp1 e
3  left outer join dept d
4    on e.deptno=d.deptno;

```

EMPNO	ENAME	JOB	DEPTNO	DNAME
7839	king	president	10	ACCOUNTING
7934	miller	clerk	10	ACCOUNTING
7369	smith	clerk	20	RESEARCH
7566	jones	manager	20	RESEARCH
7788	scott	analyst	20	RESEARCH
7499	allen	salesman	30	SALES
7698	blake	manager	30	SALES
7844	turner	salesman	30	SALES
7900	james	clerk	30	SALES

9 rows selected.

```

SQL> select ename,dname
2   from emp1 e
3  right outer join dept d
4    on e.deptno=d.deptno;

```

ENAME	DNAME
smith	RESEARCH
allen	SALES
jones	RESEARCH
blake	SALES
scott	RESEARCH
king	ACCOUNTING
turner	SALES
james	SALES
miller	ACCOUNTING
	OPERATIONS

10 rows selected.

```
SQL> select empno,dname,location
2  from emp1 e
3  full outer join dept d
4  on e.deptno=d.deptno;
```

EMPNO	DNAME	LOCATION
7369	RESEARCH	DALLAS
7499	SALES	CHICAGO
7566	RESEARCH	DALLAS
7698	SALES	CHICAGO
7788	RESEARCH	DALLAS
7839	ACCOUNTING	NEW YORK
7844	SALES	CHICAGO
7900	SALES	CHICAGO
7934	ACCOUNTING	NEW YORK
	OPERATIONS	BOSTON

```
10 rows selected.
```